The commutators of canonical transcorrelation

$\hat{C} = [\hat{F}, \hat{A}]$, $\hat{D} = [\hat{V}, \hat{A}]$, and $\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}]$

The three commutators the CT expansion needs, gathered. Each is derived in its own note — `ct_F_A.tex`, `ct_V_A.tex`, `ct_f_a_a.tex` — and each was read from a sheet of the handwritten notes, reproduced as a scan in §4. This document collects the results, the index conventions they are written in, and what has been checked.

1. Conventions

The crossed Fock order of the sheets and of `formula.tex`, excitation operators, and general indices ranging over the whole space $\{i\} \cup \{a\}$:

$$\hat{F} = f^p_q \, \hat{E}^q_p, \qquad \hat{V} = \tfrac{1}{2} \, g^{pq}_{rs} \, \hat{E}^{rs}_{pq}, \qquad \hat{T}_1 = t^a_i \, \hat{E}^a_i, \qquad \hat{T}_2 = \tfrac{1}{2} \, t^{a\beta}_{ij} \, \hat{E}^{a\beta}_{ij}, \tag{1}$$

with $\hat{A} = \hat{T} - \hat{T}^\dagger$, $\hat{T} = \hat{T}_1 + \hat{T}_2$, and g^{pq}_{rs}, $t^{a\beta}_{ij}$ antisymmetric in each of their pairs. Two points that the checks depend on:

- $\hat{F}$ is Hermitian only for symmetric (real) f, which is the physical case; the identity $[\hat{F}, \hat{A}] = [\hat{F}, \hat{T}] + [\hat{F}, \hat{T}]^\dagger$ uses that Hermiticity and fails otherwise.
- A general index is occupied or virtual and nothing else, so an occupied label can never satisfy a delta against a virtual one. That is how some terms below vanish — by an unsatisfiable delta, not by cancellation.

2. $[\hat{F}, \hat{A}]$

Both blocks come from the $(1,1)$ and $(1,2)$ rules,

$$[\hat{E}^q_p, \hat{E}^s_r] = \delta^s_p \hat{E}^q_r - \delta^q_r \hat{E}^s_p, \qquad [\hat{E}^q_p, \hat{E}^{tu}_{rs}] = \delta^t_p \hat{E}^{qu}_{rs} + \delta^u_p \hat{E}^{qt}_{sr} - \delta^q_r \hat{E}^{tu}_{ps} - \delta^q_s \hat{E}^{ut}_{pr}, \tag{2}$$

and with $[\hat{F}, \hat{A}] = [\hat{F}, \hat{T}] + [\hat{F}, \hat{T}]^\dagger$ one gets

$$[\hat{F}, \hat{T}_1] = t^a_i (f^a_q \hat{E}^q_i - f^p_i \hat{E}^a_p), \tag{3}$$

$$[\hat{F}, \hat{T}_2] = \tfrac{1}{2}\Big(+ f^a_q t^{a\beta}_{ij} \hat{E}^{q\beta}_{ij} + f^\beta_q t^{a\beta}_{ij} \hat{E}^{qa}_{ji} - f^p_i t^{a\beta}_{ij} \hat{E}^{a\beta}_{pj} - f^j_p t^{a\beta}_{ij} \hat{E}^{\beta a}_{pi} \Big), \tag{4}$$

the four terms of (4) kept separate: they are different operators, and cancelling a pair would merge coefficients belonging to different index patterns. Collecting both halves in the general basis,

$$\boxed{[\hat{F}, \hat{A}] = C^q_p \, \hat{E}^q_p + C^{rs}_{pq} \, \hat{E}^{rs}_{pq}} \tag{5}$$

with

$$\boxed{C^q_p = \underbrace{f^a_q \, t^a_p}_{p \text{ occupied}} - \underbrace{f^p_i \, t^q_i}_{q \text{ virtual}} + \underbrace{f^a_p \, t^a_q}_{q \text{ occupied}} - \underbrace{f^q_i \, t^p_i}_{p \text{ virtual}}} \tag{6}$$

for the one-body coefficient, and the eight index patterns of the four terms of (4) together with their conjugates for the two-body coefficient C^{rs}_{pq}. The conjugate half is what supplies the third and fourth terms of (6); the sheet's two-term form is the $[\hat{F},\hat{T}]$ half alone.

3. $[\hat{V},\hat{A}]$

The $(2,1)$ commutator is minus the $(1,2)$ one of (2), which gives the four-term block

$$[\hat{V},\hat{T}_1] = \tfrac{1}{2}\, g^{pq}_{rs} t^a_i \Big(\delta^a_p \hat{E}^{rs}_{iq} + \delta^a_q \hat{E}^{sr}_{ip} - \delta^r_i \hat{E}^{as}_{pq} - \delta^s_i \hat{E}^{ar}_{qp}\Big), \tag{7}$$

and the $(2,2)$ rule gives the twelve-term block

$$\begin{aligned}
[\hat{E}^{rs}_{pq}, \hat{E}^{a\beta}_{ij}] = {} & + \delta^a_p\, \hat{E}^{rs\beta}_{iqj} + \delta^\beta_p\, \hat{E}^{rsa}_{jqi} + \delta^a_q\, \hat{E}^{sr\beta}_{ipj} + \delta^\beta_q\, \hat{E}^{sra}_{jpi} \\
& + \delta^a_p \delta^\beta_q\, \hat{E}^{rs}_{ij} + \delta^\beta_p \delta^a_q\, \hat{E}^{rs}_{ji} \\
& - \delta^r_i\, \hat{E}^{a\beta s}_{pvq} - \delta^s_i\, \hat{E}^{a\beta r}_{qvp} - \delta^r_j\, \hat{E}^{\beta as}_{piq} - \delta^s_j\, \hat{E}^{\beta ar}_{qip} \\
& - \delta^r_i \delta^s_j\, \hat{E}^{a\beta}_{pq} - \delta^s_i \delta^r_j\, \hat{E}^{a\beta}_{qp},
\end{aligned} \tag{8}$$

so that $[\hat{V},\hat{T}_2]$ is $\frac{1}{4} g^{pq}_{rs} t^{a\beta}_{ij}$ times this bracket. The four terms carrying two deltas must not be dropped: with the eight single-delta terms alone the block is reproduced only to a relative 4.1×10^{-1}, with the four double-delta terms alone to 5.9×10^{-1}, and with all twelve to 3.8×10^{-16}. Contracting one index of g per delta gives the coefficient form, twelve terms, in `ct_V_A.tex` eq. (13).

4. $\frac{1}{2}[[\hat{F},\hat{A}],\hat{A}]$

With $\hat{C} = [\hat{F},\hat{A}]$ Hermitian,

$$\tfrac{1}{2}\,[[\hat{F},\hat{A}],\hat{A}] = \tfrac{1}{2}[\hat{C},\hat{T}] + \tfrac{1}{2}[\hat{C},\hat{T}]^\dagger, \qquad \hat{C} = \hat{C}_1 + \hat{C}_2 + \hat{C}_1^\dagger + \hat{C}_2^\dagger, \qquad \hat{C}_r = [\hat{F},\hat{T}_r], \tag{9}$$

which expands into the eight commutators of the sheets,

$$\tfrac{1}{2}\Big\{[\hat{C}_1,\hat{T}_1] + [\hat{C}_1,\hat{T}_2] + [\hat{C}_2,\hat{T}_1] + [\hat{C}_2,\hat{T}_2] + [\hat{C}_1^\dagger,\hat{T}_1] + [\hat{C}_1^\dagger,\hat{T}_2] + [\hat{C}_2^\dagger,\hat{T}_1] + [\hat{C}_2^\dagger,\hat{T}_2]\Big\} + \text{h.c.}. \tag{10}$$

None of the eight is redundant: $[\hat{C}_r,\hat{T}_s]^\dagger = -[\hat{C}_r^\dagger,\hat{T}_s^\dagger]$ relates conjugates across $\hat{T}_s^\dagger$, and the two mixed nested commutators differ because $\hat{T}_1\hat{T}_2 - \hat{T}_2\hat{T}_1 \neq 0$.

The simplest block shows both mechanisms at work. From (3) and $\hat{T}_1 = t^\beta_j \hat{E}^\beta_j$,

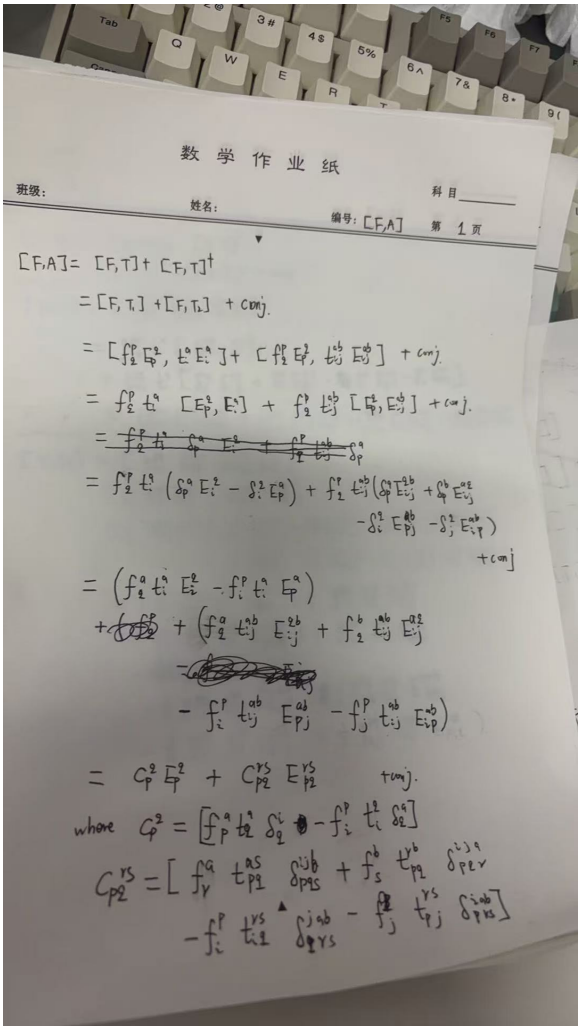
$$[\hat{E}^q_i, \hat{E}^\beta_j] = \delta^\beta_i \hat{E}^q_j - \delta^q_j \hat{E}^\beta_i, \qquad [\hat{E}^a_p, \hat{E}^\beta_j] = \delta^\beta_p \hat{E}^a_j - \delta^a_j \hat{E}^\beta_p, \tag{11}$$

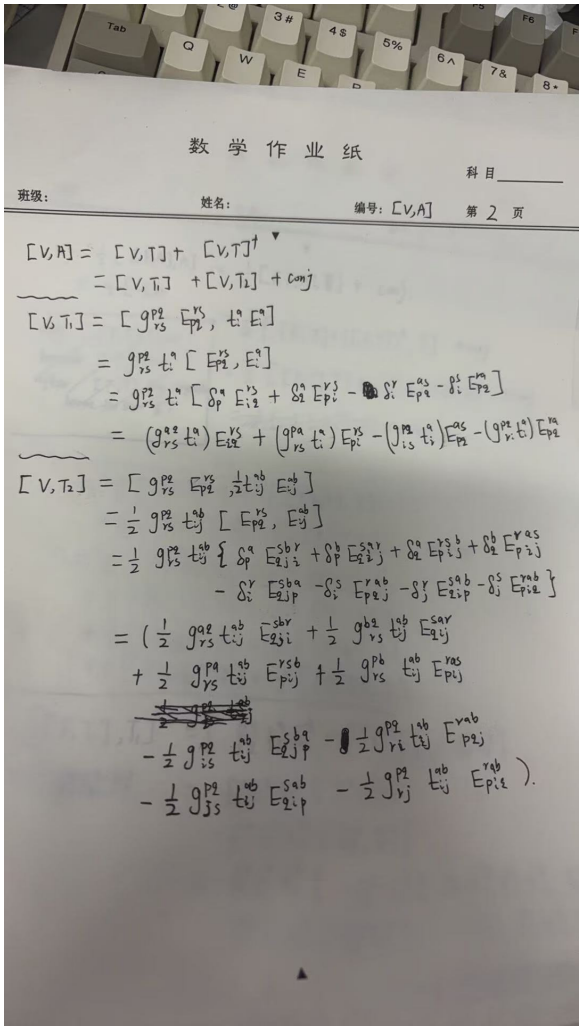
and two of the four terms vanish because δ^β_i and δ^a_j tie an occupied label to a virtual one. The surviving two are equal under $i \leftrightarrow j$, $a \leftrightarrow \beta$, so

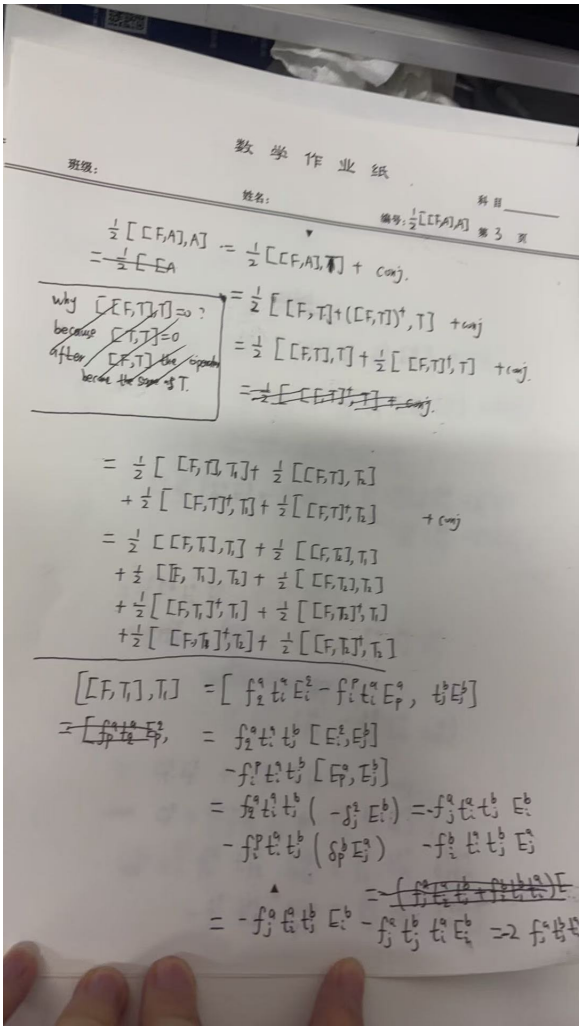
$$[[\hat{F},\hat{T}_1],\hat{T}_1] = -2\, f^a_j\, t^a_i\, t^\beta_j\, \hat{E}^\beta_i \neq 0 : \tag{12}$$

the block does not vanish, because $[\hat{F},\hat{T}_1]$ is a one-body operator of the same shape as $\hat{T}_1$ with different coefficients.

5. The sheets







Left to right, top to bottom: the $[\hat{F}, \hat{A}]$ page, the $[\hat{V}, \hat{A}]$ page, and the double-commutator page. Everything in §2–§4 was read from these, and each equation was checked against the others' machinery rather than against the handwriting.

6. What is checked

quantity	check	max$\lvert\Delta\rvert$	relative
$[\hat{F}, \hat{T}_1]$, (3)	`test_F_A_commutator.py`	1.1×10^{-15}	2.6×10^{-16}
$[\hat{F}, \hat{T}_2]$, (4), four terms	idem	3.6×10^{-15}	1.5×10^{-16}
$[\hat{F}, \hat{A}]$, (5), both coefficients	`test_F_A_result.py`	4.4×10^{-15}	6.5×10^{-16}
C^q_p, (6)	idem	—	2.5×10^{-16}
$[\hat{V}, \hat{T}_1]$, (7)	`test_V_A_commutator.py`	8.9×10^{-15}	6.1×10^{-16}
$[\hat{V}, \hat{T}_2]$, (8), twelve terms	idem	5.0×10^{-14}	8.9×10^{-16}
the eight blocks of (10)	`test_ct_structure.py`	7.4×10^{-13}	—
$[[\hat{F}, \hat{T}_1], \hat{T}_1]$, (12)	idem	2.2×10^{-15}	—

Open. The two-body coefficient C^{rs}_{pq} of (5) is verified as a collection of index patterns and has no compact closed form written down. And the three-body operators produced by the $(2, 2)$ blocks — the four single-delta terms of (8) whose operators carry three indices — still need the $[\ldots]_{1,2}$ reduction through the extended normal ordering of Kutzelnigg and Mukherjee.

References

[1] W. Kutzelnigg and D. Mukherjee, *Normal order and extended Wick theorem for a multiconfiguration reference wave function*, J. Chem. Phys. **107**, 432 (1997).

[2] T. Yanai and T. Shiozaki, *Canonical transcorrelated theory with projected Slater-type geminals*, J. Chem. Phys. **136**, 084107 (2012).